

BG96 TCP/IP

Application Note

LPWA Module Series

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About the Document

Revision History

Version	Date	Author	Description
1.0	2017-11-23	Walker HAN/ Lucifer YAN	Initial
1.1	2019-01-31	Justice HAN/ Sherlock ZHAO	1 Updated related information about the supported IPv6 feature of the module (Chapter 2.1.1, 2.1.2 and 2.1.15) 2 Added Write Command AT+QICFG="recvind",<recv_ind> to set URC format of TCP/IP buffer mode (Chapter 2.1.15)
1.2	2020-05-28	Adonis CHEN/ Terrence YANG	1. Updated the flowchart of using TCP/IP AT commands (Figure 1). 2. Added the following commands: ● AT+QICFG="tcp/keepalive",<keepalive>,<idle_time>,<interval_time>,<probe_cnt> ● AT+QICFG="ipv6prior",<IPv6_prior> 3. Added examples relating to TCP/IP operations under IPv6 (Chapter 3.3, 3.7 and 3.9).

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1 Introduction

Quectel BG96 module features embedded TCP/IP stack, which enables the host to access the Internet directly over AT commands. This greatly reduces the dependence on the PPP and external TCP/IP protocol stacks and thus minimizes the cost.

Quectel BG96 module provides the following socket services:

- TCP client
- UDP client
- TCP server
- UDP server

1.1. The Process of Using TCP/IP AT Commands

Through TCP/IP AT commands, the host can configure a PDP context, activate/deactivate the PDP context, start/close socket service and send/receive data via socket service. The following figure illustrates how to use TCP/IP AT commands.

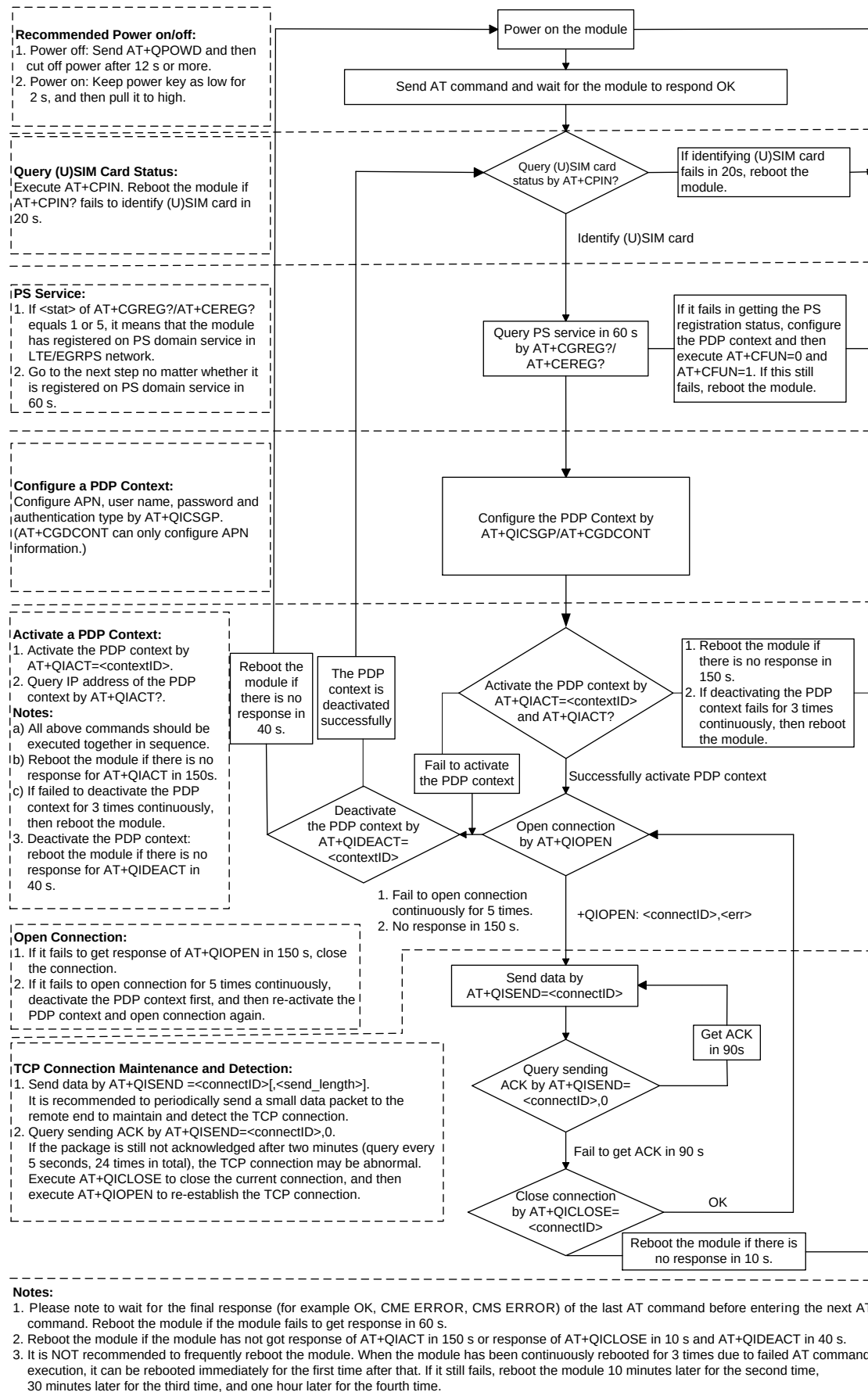


Figure 1: Flow Chart of Using TCP/IP AT Commands

1.2. Description of Data Access Modes

BG96 module supports the following three data access modes:

- Buffer access mode
- Direct push mode
- Transparent access mode

When opening a socket service via **AT+QIOPEN**, the data access mode can be specified by **<access_mode>**. After the socket service is opened, **AT+QISWTMD** allows switching between the access modes as needed.

1. In buffer access mode, data can be sent via **AT+QISEND**, and if the module has received the data from the Internet, it will buffer the data and report the URC **+QIURC: "recv",<connectID>**. The received data can be read via **AT+QIRD**.
2. In direct push mode, data can be sent via **AT+QISEND**, and if the module has received the data from the Internet, the data will be outputted to COM port directly in the following format:
+QIURC: "recv",<connectID>,<currentrecvlength><CR><LF><data>.
3. In transparent access mode, the corresponding port (such as UART, USB modem port, etc.) enters exclusive mode. The data received from COM port will be sent to the Internet directly, and the data received from the Internet will be outputted via COM port directly. **+++** can be used to exit from transparent access mode. When **OK** is returned, the module will be switched to buffer access mode. **AT+QISWTMD** can be used to switch back to transparent access mode.

- **Exit from transparent access mode**

To exit from the transparent access mode, **+++** or DTR (**AT&D1** should be set first) can be used. To prevent the **+++** from being misinterpreted as data, the following sequence should be followed:

- 1) Do not input any character within 1 s or longer before inputting **+++**.
- 2) Input **+++** within 1 s, and no other characters should be inputted during the time.
- 3) Do not input any character within 1 s after **+++** has been inputted.
- 4) Use **+++** or DTR (**AT&D1** should be set first) to make the module exit from transparent access mode, and wait until **OK** is returned.

- **Return back to transparent access mode**

- 1) By **AT+QISWTMD**. Specify **<access_mode>** as 2 when executing this command. When transparent access mode is entered successfully, **CONNECT** will be returned.
- 2) By **ATO**. After a connection exits from transparent access mode, executing **ATO** will switch the data access mode back to transparent access mode again. When transparent access mode is entered successfully, **CONNECT** will be returned. If no connection enters transparent access mode before, **ATO** will return **NO CARRIER**.

NOTES

1. In buffer access mode, if the buffer is not empty, the module will not report a new URC until all the received data has been read via **AT+QIRD** from buffer.
2. In transparent access mode, AT commands cannot be executed. If the socket connection is closed because of network error or other errors, the module will report **NO CARRIER** and exit from the transparent access mode. In this case, **AT+QICLOSE** should be executed to close the socket service.

2 TCP/IP AT Commands

2.1. AT Command Syntax

2.1.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on command line. When an optional parameter is omitted, the new value equals its previous value or its default setting, unless otherwise specified.
- **Underline** Default setting of a parameter.

2.1.2. AT Command Syntax

The **AT** or **at** prefix must be added at the beginning of each command line. Entering **<CR>** will terminate a command line. Commands are usually followed by a response that includes **<CR><LF><response><CR><LF>**. Throughout this document, only the response **<response>** will be presented, **<CR><LF>** are omitted intentionally.

Table 1: Type of AT Commands and Responses

Test Command	AT+<cmd>=?	This command returns the list of parameters and value ranges set by the corresponding Write Command or internal processes.
Read Command	AT+<cmd>?	This command returns the currently set value of the parameter or parameters.
Write Command	AT+<cmd>=<p1> [,<p2>[,<p3>[...]]]	This command sets the user-definable parameter values.
Execution Command	AT+<cmd>	This command reads non-variable parameters affected by internal processes in the module.

2.2. Description of TCP/IP AT Commands

2.2.1. AT+QICSGP Configure Parameters of a TCP/IP Context

The Write Command configures <APN>, <username>, <password> and other parameters of a TCP/IP context.

AT+QICSGP Configure Parameters of a TCP/IP Context	
Test Command AT+QICSGP=?	Response +QICSGP: (range of supported <contextID>s),(range of supported <context_type>s),<APN>,<username>,<password>,(range of supported <authentication>s) OK
Write Command Query/configure a specified context AT+QICSGP=<contextID>[,<context_type>,<APN>,<username>,<password>][,<authentication>]]	Response If the optional parameters are omitted, query the current setting: +QICSGP: <context_type>,<APN>,<username>,<password>,<authentication> OK If any of the optional parameters is specified, configure a specified context: OK If there is any error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately (actually determined by network). The configurations will be saved automatically.

Parameter

<contextID>	Integer type. Context ID. Range: 1–16.
<context_type>	Integer type. Protocol type. 1 IPv4 2 IPv6 3 IPv4v6
<APN>	String type. Access point name.
<username>	String type. Username.

<password>	String type. Password.
<authentication>	Integer type. Authentication methods.
0	None
1	PAP
2	CHAP
3	PAP or CHAP

Example

```

AT+QICSGP=1 //Query the configuration of context 1.
+QICSGP: 1,"", "", "", 0

OK
AT+QICSGP=1,1,"UNINET","", "", 1 //Configure context 1. APN is "UNINET" for China Unicom.
OK

```

2.2.2. AT+QIACT Activate a PDP Context

Before activating a PDP context via **AT+QIACT**, the context should be configured by **AT+QICSGP**. After activation, the IP address can be queried via **AT+QIACT?**. Although the range of <contextID> is 1–16, the module supports maximum three PDP contexts activated simultaneously under LTE Cat M/EGPRS and maximum two under LTE Cat NB1. Depending on the network, it may take at most 150 seconds to return **OK** or **ERROR** after executing **AT+QIACT**. Before the response is returned, other AT commands cannot be executed.

AT+QIACT Activate a PDP Context

Test Command AT+QIACT=?	Response +QIACT: (range of supported <contextID>s) OK
Read Command AT+QIACT?	Response Return the list of the current activated contexts and their IP addresses: +QIACT: 1,<context_state>,<context_type>,<IP_address> [..... +QIACT: 16,<context_state>,<context_type>,<IP_addresses>] OK
Write Command Activate a specified PDP context AT+QIACT=<contextID>	Response OK

	If there is any error: ERROR
Maximum Response Time	150 s, determined by network.
Characteristics	/

Parameter

<contextID>	Integer type. Context ID. Range: 1–16.
<context_state>	Integer type. Context state. 0 Deactivated 1 Activated
<context_type>	Integer type. Protocol type. <u>1</u> IPv4 2 IPv6
<IP_address>	String type. The local IP address after the context is activated.

2.2.3. AT+QIDEACT Deactivate a PDP Context

The Write Command deactivates a specific context and close all TCP/IP connections set up in this context. Depending on the network, it may take at most 40 seconds to return **OK** or **ERROR** after executing **AT+QIDEACT**. Before the response is returned, other AT commands cannot be executed.

AT+QIDEACT Deactivate a PDP Context

Test Command AT+QIDEACT=?	Response +QIDEACT: (range of supported <contextID>s) OK
Write Command AT+QIDEACT=<contextID>	Response OK If there is any error: ERROR
Maximum Response Time	40 s, determined by network.
Characteristics	/

Parameter

<contextID>	Integer type. Context ID. Range: 1–16.
-------------	--

2.2.4. AT+QIOPEN Open a Socket Service

The Write Command opens a socket service. The service type can be specified by <service_type>. The data access mode (buffer access mode, direct push mode and transparent access mode) can be specified by <access_mode>. URC **+QIOPEN** indicates whether the socket service has been opened successfully.

- If <service_type> is "TCP LISTENER", the module works as a TCP server. After accepting a new TCP connection, the module will automatically specify a <connectID> and report URC **+QIURC: "incoming",<connectID>,<serverID>,<remoteIP>,<remote_port>**. The range of <connectID> is 0–11. The type of this new incoming connection is "TCP INCOMING" and the <access_mode> of "TCP INCOMING" is as same as the <access_mode> of "TCP LISTENER".
- If <service_type> is "UDP SERVICE", UDP data can be sent to or received from the remote IP via <local_port>.
 - Send data: execute **AT+QISEND=<connectID>,<send_length>,<remoteIP>,<remote_port>**.
 - Receive data in direct push mode: the module reports URC **+QIURC: "recv",<connectID>,<currentrecvlength>,<remoteIP>,<remote_port><CR><LF><data>**.
 - Receive data in buffer access mode: the module reports URC **+QIURC: "recv",<connectID>**, and then the data can be retrieved via **AT+QIRD=<connectID>**.
- It is suggested to wait for 150 seconds for **+QIOPEN: <connectID>,<err>** to be outputted. If the URC cannot be received in 150 seconds, **AT+QICLOSE** should be used to close the socket.

AT+QIOPEN Open a Socket Service

Test Command AT+QIOPEN=?	Response +QIOPEN: (range of supported <contextID>s),(range of supported <connectID>s),"TCP/UDP/TCP LISTENER/UDP SERVICE",<IP_address>/<domain_name>,(range of supported <remote_port>s),(range of supported <local_port>s),(range of supported <access_mode>s) OK
Write Command AT+QIOPEN=<contextID>,<connectID>,<service_type>,<IP_address>/<domain_name>,<remote_port>[,<local_port>[,<access_mode>]]	Response If <access_mode>=2 and the socket service is opened successfully: CONNECT

	If there is any error: ERROR Error description can be got via AT+QIGETERROR. If <access_mode>=0 or 1: OK +QIOPEN: <connectID>,<err> <err> is 0 when the service is opened successfully. In other cases, <err> is not 0.
Maximum Response Time	300 s, determined by network.
Characteristics	/

Parameter

<contextID>	Integer type. Context ID. Range: 1–16.
<connectID>	Integer type. Socket service index. Range: 0–11.
<service_type>	String type. Socket service type. "TCP" Start a TCP connection as a client "UDP" Start a UDP connection as a client "TCP LISTENER" Start a TCP server to listen to TCP connection "UDP SERVICE" Start a UDP service
<IP_address>	String type. If <service_type> is "TCP" or "UDP", it indicates the IP address of a remote server, such as "220.180.239.212". If <service_type> is "TCP LISTENER" or "UDP SERVICE", enter "127.0.0.1".
<domain_name>	String type. Domain name address of the remote server.
<remote_port>	Integer type. Port number of the remote server. It is only valid when <service_type> is "TCP" or "UDP", and in such as case, the range is 0–65535. Otherwise, any value in the range of 0–65535 will not take effect.
<local_port>	Integer type. Local port number. Range: 0–65535. If <service_type> is "TCP LISTENER" or "UDP SERVICE", this parameter must be specified. If <service_type> is "TCP" or "UDP", <local_port> will be 0, then the local port will be assigned automatically. In this case, the value 0 will be used if it is omitted.
<access_mode>	Integer type. Data access mode of the socket service. 0 Buffer access mode 1 Direct push mode 2 Transparent access mode
<err>	Integer type. Error code of operation. Please refer to Chapter 4 .

2.2.5. AT+QICLOSE Close a Socket Service

The Write Command closes a specified socket service. Depending on the network, it will take at most 10 seconds (default value, can be configured by **<timeout>**) to return **OK** or **ERROR** after executing **AT+QICLOSE**. Before the response is returned, other AT commands cannot be executed.

AT+QICLOSE Close a Socket Service

Test Command AT+QICLOSE=?	Response +QICLOSE: (range of supported <connectID> s),(range of supported <timeout> s) OK
Write Command AT+QICLOSE=<connectID>[,<timeout>]	Response If the socket service is closed successfully: OK If failed to close: ERROR
Maximum Response Time	120 s
Characteristics	/

Parameter

<connectID>	Integer type. Socket service index. Range: 0–11.
<timeout>	Integer type. If the FIN ACK of the other peers is not received within <timeout> , the module will force to close the socket. Range: 0-65535. Default value: 10. Unit: second.

2.2.6. AT+QISTATE Query Socket Service Status

The Write Command queries the socket service status. If **<query_type>=0**, it will return the status of all existing socket services in the specified context. If **<query_type>=1**, it will return the status of a specified socket service.

AT+QISTATE Query Socket Service Status

Test Command AT+QISTATE=?	Response OK
Read/Execution Command AT+QISTATE? or AT+QISTATE	Response Return the status of all existing connections: +QISTATE: <connectID> , <service_type> , <IP_address> , <r

	emote_port>,<local_port>,<socket_state>,<contextID>,<serverID>,<access_mode>,<AT_port> [...] OK
Write Command If <query_type> is 0, query the connection status of a specified context AT+QISTATE=<query_type>,<context ID>	Response Return the status of all existing connections in a specified context: +QISTATE: <connectID>,<service_type>,<IP_address>,<remote_port>,<local_port>,<socket_state>,<contextID>,<serverID>,<access_mode>,<AT_port> [...] OK
Write Command If <query_type> is 1, query the connection status of a specified socket service AT+QISTATE=<query_type>,<connect ID>	Response +QISTATE: <connectID>,<service_type>,<IP_address>,<remote_port>,<local_port>,<socket_state>,<contextID>,<serverID>,<access_mode>,<AT_port> OK
Maximum Response Time	300 ms
Characteristics	/

Parameter

<query_type>	Integer type. Query type. 0 Query connection status of all socket services in a specified context 1 Query connection status of a specified socket service
<contextID>	Integer type. Context ID. Range: 1–16.
<connectID>	Integer type. Socket service index. Range: 0–11.
<service_type>	String type. Socket service type. "TCP" Start a TCP connection as a client "UDP" Start a UDP connection as a client "TCP LISTENER" Start a TCP server to listen to TCP connection "TCP INCOMING" Start a TCP connection accepted by a TCP server "UDP SERVICE" Start a UDP service
<IP_address>	String type. IP address. If <service_type> ="TCP" or "UDP", it is the IP address of remote server. If <service_type> ="TCP LISTENER" or "UDP SERVICE", it is the local IP address. If <service_type> ="TCP INCOMING", it is the IP address of remote client.
<remote_port>	Integer type. Remote port number.

	If <service_type> ="TCP" or "UDP", it is the port of remote server.
	If <service_type> ="TCP LISTENER" or "UDP SERVICE", the port is invalid.
	If <service_type> ="TCP INCOMING", it is the port of remote client.
<local_port>	Integer type. Local port number. Range: 0–65535. If <local_port> is 0, then the local port is assigned automatically.
<socket_state>	Integer type. Socket service status.
	0 "Initial": connection has not been established
	1 "Opening": client is connecting or server is trying to listen
	2 "Connected": client/incoming connection has been established
	3 "Listening": server is listening
	4 "Closing": connection is closing
<serverID>	Integer type. Only valid when <service_type> is "TCP INCOMING". <serverID> represents which server accepts this TCP incoming connection, and the value is the same as <connectID> of this server's "TCP LISTENER".
<access_mode>	Integer type. Data access mode.
	0 Buffer access mode
	1 Direct push mode
	2 Transparent access mode
<AT_port>	String type. COM port of socket service.
	"usbmodem" USB modem port
	"usbat" USB AT port
	"uart1" UART1 port

2.2.7. AT+QISEND Send Data

In buffer access mode (**<access_mode>**=0) or direct push mode (**<access_mode>**=1), the data can be sent via **AT+QISEND**. If the data is sent to the module successfully, **SEND OK** will be returned. Otherwise it will return **SEND FAIL** or **ERROR**. **SEND FAIL** indicates the sending buffer is full and the data can be tried to be resent. **ERROR** indicates it encounters an error in the process of sending data and the data should be delayed for some time to be sent. **SEND OK** does not mean the data has been sent to the server successfully. Please query whether the data has reached the server by **AT+QISEND=<connectID>,0**.

AT+QISEND Send Data

Test Command AT+QISEND=?	Response +QISEND: (range of supported <connectID> s),(range of supported <send_length> s) OK
Write Command Send variable-length data when <service_type> is "TCP", "UDP" or "TCP INCOMING"	Response > After the response >, input the data to be sent. Tap Ctrl+Z to send the data, and tap Esc to cancel the operation.

AT+QISEND=<connectID>	If the connection is established and the data is sent successfully: SEND OK If the connection is established but the sending buffer is full: SEND FAIL If the connection is not established, abnormally closed, or any parameter is incorrect: ERROR
Write Command Send fixed-length data when <service_type> is "TCP", "UDP" or "TCP INCOMING" AT+QISEND=<connectID>,<send_length>	Response > After the response >, input the data until the data length is equal to <send_length>. If the connection is established and the data is sent successfully: SEND OK If the connection is established but the sending buffer is full: SEND FAIL If the connection is not established, abnormally closed, or any parameter is incorrect: ERROR
Write Command When <service_type> is "UDP SERVICE" AT+QISEND=<connectID>,<send_length>,<remoteIP>,<remote_port>	Response This command is used to send fixed-length data to specified remote IP address and remote port. <service_type> must be "UDP SERVICE". > After the response >, input the data until the data length is equal to <send_length> If the connection is established and the data is sent successfully: SEND OK If the connection is established but the sending buffer is fully: SEND FAIL If the connection is not established, abnormally closed, or any parameter is incorrect: ERROR

Write Command When <send_length> is 0, query the sent data AT+QISEND=<connectID>,0	Response If the specified <connectID> connection exists: +QISEND: <total_send_length>,<ackedbytes>,<unackedbytes> OK If there is any error: ERROR
Maximum Response Time	120 s, determined by network.
Characteristics	/

Parameter

<connectID>	Integer type. Socket service index. Range: 0–11.
<send_length>	Integer type. Length of the data to be sent, which cannot exceed 1460 bytes.
<remoteIP>	String type. Remote IP address (must be dot format). It is only valid when <service_type> is "UDP SERVICE".
<remote_port>	Integer type. Remote port. It is only valid when <service_type> is "UDP SERVICE".
<total_send_length>	Integer type. Total length of sent data. Unit: byte.
<ackedbytes>	Integer type. Total length of acknowledged data. Unit: byte.
<unackedbytes>	Integer type. Total length of unacknowledged data. Unit: byte.
<err>	Integer type. Error code of operation. Please refer to Chapter 4 .

2.2.8. AT+QIRD Retrieve the Received TCP/IP Data

In buffer access mode, after receiving data, the module will buffer it and report **+QIURC: "recv",<connectID>**, then the data can be retrieved by **AT+QIRD**.

Please note that if the buffer is not empty, and the module receives data again, it will not report a new URC until all the received data are retrieved via **AT+QIRD** from buffer.

AT+QIRD Retrieve the Received TCP/IP Data

Test Command AT+QIRD=?	Response +QIRD: (range of supported <connectID>s),(range of supported <read_length>s) OK
Write Command When <service_type> is "TCP" or	Response If the specified connection has received data:

"UDP" or "TCP INCOMING" AT+QIRD=<connectID>[,<read_length>]	+QIRD: <read_actual_length><CR><LF><data> OK If there is no data: +QIRD: 0 OK If the connection does not exist: ERROR
Write Command When <service_type> is "UDP SERVICE" AT+QIRD=<connectID>	Response If the data exists: +QIRD: <read_actual_length>,<remoteIP>,<remote_port><CR><LF><data> OK If there is no data: +QIRD: 0 OK If the connection does not exist: ERROR
Write Command When <read_length> is 0, query the retrieved data length AT+QIRD=<connectID>,0	Response If the specified connection exists: +QIRD: <total_receive_length>,<have_read_length>,<unread_length> OK If there is any error: ERROR
Maximum Response Time	120 s, determined by network.
Characteristics	/

Parameter

<connectID>	Integer type. Socket service index. Range: 0–11.
<read_length>	Integer type. Maximum length of the data to be retrieved. Range: 0-1500.

	Default value: 1500. Unit: byte.
<read_actual_length>	Integer type. Length of the data that has been actually retrieved. Unit: byte.
<remoteIP>	String type. Remote IP address. It is only valid when <service_type> is "UDP SERVICE".
<remote_port>	Integer type. Remote port. It is only valid when <service_type> is "UDP SERVICE".
<data>	String type. Retrieved data.
<total_receive_length>	Integer type. Total length of received data. Unit: byte.
<have_read_length>	Integer type. Length of the data that has been retrieved. Unit: byte.
<unread_length>	Integer type. Length of the data that has not been retrieved. Unit: byte.

2.2.9. AT+QISENDEX Send Hex String Data

The Write Command sends hex string data and cannot be applied for "UDP SERVICE" and "TCP LISTENER" sockets.

AT+QISENDEX Send Hex String Data

Test Command AT+QISENDEX=?	Response +QISENDEX: (range of supported <connectID>s),<hex_string> OK
Write Command AT+QISENDEX=<connectID>,<hex_string>	Response If the hex string data is sent successfully: SEND OK If the sending buffer is full: SEND FAIL If the connection does not exist: ERROR
Maximum Response Time	120 s, determined by network.
Characteristics	/

Parameter

<connectID>	Integer type. Socket service index. Range: 0–11.
<hex_string>	String type. Hex string data. The maximum length is 512 bytes.

2.2.10. AT+QISWTMD Switch Data Access Mode

The Write Command switches data access mode between buffer access mode, direct push mode and transparent access mode. When starting a socket service, the data access mode can be specified by **<access_mode>** of **AT+QIOPEN**. After the socket is opened, the data access mode can be changed via **AT+QISWTMD**.

AT+QISWTMD Switch Data Access Mode

Test Command AT+QISWTMD=?	Response +QISWTMD: (range of supported <connectID>s),(range of supported <access_mode>s) OK
Write Command AT+QISWTMD=<connectID>,<access_mode>	Response If the data access mode is switched successfully and <access_mode> is 0 or 1: OK If the data access mode is switched successfully and <access_mode> is 2, the module enters data mode: CONNECT If there is any error: ERROR
Maximum Response Time	300 ms
Characteristics	/

Parameter

<connectID>	Integer type. Socket service index. Range: 0–11.
<access_mode>	Integer type. Data access modes of the connection.
0	Buffer access mode
1	Direct push mode
2	Transparent access mode

2.2.11. AT+QPING Ping a Remote Server

The Write Command tests the Internet protocol reachability of a host. Before using ping tools, the host should activate the context corresponding to **<contextID>** via **AT+QIACT** first. It will return the result within **<timeout>** and the default value of **<timeout>** is 4.

AT+QPING Ping a Remote Server

Test Command AT+QPING=?	Response +QPING: (range of supported <connectID>s),<host>,(range of supported <timeout>s),(range of supported <pingnum>s) OK
Write Command AT+QPING=<contextID>,<host>,<timeout>,<pingnum>]	Response If a remote server is pinged successfully: OK +QPING: <result>[,<IP_address>,<bytes>,<time>,<ttl>]<CR><LF>[...] +QPING: <finresult>[,<sent>,<rcvd>,<lost>,<min>,<max>,<avg>] If there is any error: ERROR
Maximum Response Time	255s, determined by <timeout>.
Characteristics	/

Parameter

<contextID>	Integer type. Context ID. Range: 1–16.
<host>	String type. Host address. The format is a domain name or a dotted decimal IP address.
<timeout>	Integer type. Maximum time to wait for the response of each ping request. Range: 1–255. Default value: 4. Unit: second.
<pingnum>	Integer type. Maximum number of times for sending ping request. Range: 1–10. Default value: 4.
<result>	Integer type. Result of each ping request. 0 Received the ping response from the server. In this case, it is followed by ,<IP_address>,<bytes>,<time>,<ttl>. Others Error code of operation. Please refer to the error codes in Chapter 4 .
<IP_address>	String type. IP address of the remote server formatted as a dotted decimal IP.
<bytes>	Integer type. Length of each sent ping request. Unit: byte.
<time>	Integer type. Time wait for the response of the ping request. Unit: ms.
<ttl>	Integer type. Time to live value of the response packet for the ping request.
<finresult>	Integer type. Final result of the command. 0 It is finished normally. It is successful to activate the context and find the host. In this case, it is followed by <sent>, <rcvd>, <lost>, <min>, <max> and <avg>.

	Others	Error code of operation. Please refer to the error codes in Chapter 4 .
<sent>	Integer type.	Total number of sending the ping requests.
<rcvd>	Integer type.	Total number of the ping requests that received the response.
<lost>	Integer type.	Total number of the ping requests that are timeout.
<min>	Integer type.	Minimum response time. Unit: ms.
<max>	Integer type.	Maximum response time. Unit: ms.
<avg>	Integer type.	Average response time. Unit: ms.

2.2.12. AT+QNTTP Synchronize Local Time with NTP Server

The Write Command synchronizes UTC with the NTP server. Before using NTP, the host should activate the context corresponding to <contextID> via **AT+QIAC** first. Depending on the network, it will take at most 125 seconds to return the result.

AT+QNTTP Synchronize Local Time with NTP Server

Test Command AT+QNTTP=?	Response +QNTTP: (range of supported <connectID>s),<server>,(range of supported <port>s),(list of supported <auto_set_time>s) OK
Read Command AT+QNTTP?	Response If in the process of local time synchronization: +QNTTP: <server>,<port> OK
Write Command AT+QNTTP=<contextID>,<server>[,<port>[,<auto_set_time>]]	Response If the local time is synchronized with NTP server successfully: OK +QNTTP: <err>,<time> If there is any error: ERROR
Maximum Response Time	125 s, determined by network.
Characteristics	The command takes effect immediately (actually determined by network). The configurations will not be saved.

Parameter

<contextID>	Integer type. Context ID. Range: 1–16.
<server>	String type. Address of NTP server.

<port>	Integer type. Port number of NTP server. Range: 1-65535. Default value: 123.
<auto_set_time>	Integer type. Whether to automatically set synchronized time to local time.
	0 Not set
	<u>1</u> Set
<err>	Integer type. Error code of operation. Please refer to Chapter 4 .
<time>	String type. Time synchronized from NTP server.
	The format is "YYYY/MM/DD,hh:mm:ss±zz". Range of "zz": -48 to 56.

2.2.13. AT+QIDNSCFG Configure Address of DNS Server

Before setting the DNS address, the host must activate the context corresponding to <contextID> via **AT+QIACT** first.

AT+QIDNSCFG Configure Address of DNS Server	
Test command AT+QIDNSCFG=?	Response +QIDNSCFG: (range of supported <connectID>s),<pridnsaddr>,<secdnsaddr> OK
Write Command AT+QIDNSCFG=<contextID>,[<pridnsaddr>,<secdnsaddr>]	Response If the optional parameters are omitted, query the current DNS server addresses of a specified PDP context: +QIDNSCFG: <contextID>,<pridnsaddr>,<secdnsaddr> OK If any of the optional parameters is specified, set the primary and secondary DNS server addresses of a specified PDP context: OK If there is any error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations will not be saved.

Parameter

<contextID>	Integer type. PDP context ID. Range: 1–16.
<pridnsaddr>	String type. Primary DNS server address.
<secdnsaddr>	String type. Secondary DNS server address.

2.2.14. AT+QIDNSGIP Get IP Address by Domain Name

Before querying the DNS, the host should activate the context corresponding to **<contextID>** via **AT+QIACT** first. Depending on the network, it will take at most 60 seconds to return the result.

AT+QIDNSGIP Get IP Address by Domain Name	
Test Command AT+QIDNSGIP=?	Response +QIDNSGIP: (range of supported <connectID> s), <host_name> OK
Write Command AT+QIDNSGIP=<contextID>,<host_name>	Response OK +QIURC: "dnsgip", <err> , <IP_count> , <DNS_ttl> [..... +QIURC: "dnsgip", <host_IP_addr>] If there is any error: ERROR
Maximum Response Time	60 s, determined by network.
Characteristics	/

Parameter

<contextID>	Integer type. PDP context ID. Range: 1–16.
<host_name>	String type. Domain name.
<err>	Integer type. Error code of operation. Please refer to Chapter 4 .
<IP_count>	Integer type. Number of the IP addresses corresponding to <host_name> .
<DNS_ttl>	Integer type. Time to live of the DNS.
<host_IP_addr>	String type. IP address of <host_name> .

2.2.15. AT+QICFG Configure Optional Parameters

The Write Command configures some optional parameters such as the packet size for transparent access mode, the format of the data to be sent or received, and the output format of received data.

AT+QICFG Configure Optional Parameters	
Test Command AT+QICFG=?	Response +QICFG: "transpktsize",(range of supported <transpkt_size> s) +QICFG: "transwaittm",(range of supported <transwaittm> s)

	+QICFG: "dataformat",(list of supported <send_data_format>s),(list of supported <recv_data_format>s) +QICFG: "viewmode",(list of supported <view_mode>s) +QICFG: "recvind",(list of supported <recv_ind>s) +QICFG: "prefix",<IPv6_prefix>,(list of supported <prefix_length>s) +QICFG: "tcp/retranscfg",(range of supported <max_backoff_s>s),(range of supported <max_rto>s) +QICFG: "tcp/keepalive",(list of supported <keepalive>s),(range of supported <idle_time>s),(range of supported <interval_time>s),(range of supported <probe_cnt>s) +QICFG: "ipv6prior",(list of supported <IPv6_prior>s) OK
Write Command Query/set the packet size for transparent access mode AT+QICFG="transpktsize"[,<transpkt_size>]	Response If the optional parameter is omitted, query the current setting: +QICFG: "transpktsize",<transpkt_size> OK If the optional parameter is specified, set the packet size for transparent access mode: OK If there is any error: ERROR
Write Command Query/set the waiting time for transparent access mode AT+QICFG="transwaittm"[,<transwaittm>]	Response If the optional parameter is omitted, query the current setting: +QICFG: "transwaittm",<transwaittm> OK If the optional parameter is specified, set the waiting time for transparent access mode: OK If there is any error: ERROR
Write Command Query/set the format of the data to be sent and received (only for buffer access mode and direct push mode) AT+QICFG="dataformat"[,<send	Response If the optional parameters are omitted, query the current setting: +QICFG: "dataformat",<send_data_format>,<recv_data_format> OK

_data_format>,<recv_data_format>]	If the optional parameters are specified, set the format for the data to be sent and received: OK If there is any error: ERROR
Write Command Query/set the output format of received data (only for buffer access mode and direct push mode) AT+QICFG="viewmode",<view_mode>]	Response If the optional parameter is omitted, query the current setting: +QICFG: "viewmode",<view_mode> OK If the optional parameter is specified, set the output format of received data: OK If there is any error: ERROR
Write Command Query/set the URC format of TCP/IP buffer access mode. If this command is set, when the module receives data from a remote server, the data length will be added to the URC. AT+QICFG="recvind",<recv_ind>]	Response If the optional parameter is omitted, query the current setting: +QICFG: "recvind",<recv_ind> OK If the optional parameter is specified, set the URC format of TCP/IP buffer mode: OK If there is any error: ERROR
Write Command When module accesses to IPv4 address of a TCP/UDP server under IPv6 network, the module will translate IPv4 address to IPv4-embedded IPv6 address using the network-specific prefixes or the well-known prefix. AT+QICFG="prefix",<IPv6_prefix>,<prefix_length>]	Response If the optional parameters are omitted, query the current setting: +QICFG: "prefix",<IPv6_prefix>,<prefix_length> OK If the optional parameters are specified, set the network-specific prefixes or the well-known prefix: OK If there is any error: ERROR
Write Command Query/set the maximum interval	Response If the optional parameters are omitted, query the current setting:

and the number of TCP retransmission AT+QICFG="tcp/retranscfg",<max_backoffs>,<max_rto>	+QICFG: "tcp/retranscfg",<max_backoffs>,<max_rto> OK If the optional parameters are specified, set the maximum interval and the number of TCP retransmission: OK If there is any error: ERROR
Write Command Query/set whether to enable TCP keepalive feature and set the value of relevant parameters AT+QICFG="tcp/keepalive",<keepalive>,<idle_time>,<interval_time>,<probe_cnt>]]	Response If the optional parameters are omitted, query the current setting: +QICFG: "tcp/keepalive",<keepalive>,<idle_time>,<interval_time>,<probe_cnt> OK If any of the optional parameters is specified, set whether to enable TCP keepalive feature and set the value of relevant parameters: OK If there is any error: ERROR
Write Command Query/set the prior PDP type when the protocol type is IPv4v6 AT+QICFG="ipv6prior",<IPv6_prior>	Response If the optional parameter is omitted, query the current setting: +QICFG: "ipv6prior",<IPv6_prior> OK If the optional parameter is specified, set IPv4 or IPv6 as the default PDP type when the protocol type is IPv4v6 OK If there is any error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration(s) will not be saved. Please refer to the note below for special cases.

Parameter

<transpkt_size>	Integer type. Maximum length of the data packet to be sent. Range:1-1460. Default value: 1024. Unit: byte.
<transwaittm>	Integer type. In transparent access mode, if the length of data received from the port is less than the value specified by <transpkt_size>, after exceeding the time specified by <transwaittm>, the data will be sent directly. Range: 0-20. Default value: 2. Unit: 100 ms.
<send_data_format>	Integer type. Format of the data to be sent. 0 Text mode 1 Hex mode
<recv_data_format>	Integer type. Format of the data to be received. 0 Text mode 1 Hex mode
<view_mode>	Integer type. Output format of received data. 0 Output format: data header\r\n\data. 1 Output format: data header,data.
<recv_ind>	Integer type. 0 URC format without data length. URC format: +QIURC: "recv",<connectID> 1 URC format with data length. URC format: +QIURC: "recv",<connectID>,<data_len>
<IPv6_prefix>	String type. Network-specific prefixes or a well-known prefix.
<prefix_length>	Integer type. Prefix length. The values are 32, 40, 48, 56, 64 and 96. Unit: bit.
<max_backoffs>	Integer type. Maximum number of TCP retransmission. Range: 3-20. Default value: 12.
<max_rto>	Integer type. Maximum interval of TCP retransmission. Range: 5-1000. Default value: 600. Unit: 100 ms.
<keepalive>	Integer type. Whether to enable TCP keepalive feature. When it is 0, <idle_time>, <interval_time> and <probe_cnt> should be ignored. 0 Disable TCP keepalive feature 1 Enable TCP keepalive feature
<idle_time>	Integer type. Default keepalive time. Range: 1-120. Unit: minute.
<interval_time>	Integer type. Time between keepalive probes. Range: 25-100. Unit: second.
<probe_cnt>	Integer type. Maximum number of keepalive probes. Range: 3-10.
<IPv6_prior>	Integer type. 0 Set IPv4 as the prior PDP type 1 Set IPv6 as the prior PDP type

NOTE

The configurations of **AT+QICFG="tcp/retranscfg",<max_backoffs>,<max_rto>** and **AT+QICFG="tcp/keepalive",<keepalive>,<idle_time>,<interval_time>,<probe_cnt>** take effect to all TCP sockets and cannot be saved to NVM.

2.2.16. AT+QISDE Control Whether to Echo the Data for AT+QISEND

The Write Command controls whether to echo the data for **AT+QISEND**.

AT+QISDE Control Whether to Echo the Data for AT+QISEND

Test Command AT+QISDE=?	Response +QISDE: (list of supported <echo>s) OK
Read Command AT+QISDE?	Response +QISDE: <echo> OK
Write Command AT+QISDE=<echo>	Response OK If there is any error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration will not be saved.

Parameter

<echo>	Integer type. Whether to echo the data for AT+QISEND
0	Not echo the data
<u>1</u>	Echo the data

2.2.17. AT+QIGETERROR Query the Last Error Code

If **<err>** is returned after executing TCP/IP commands, the details of error can be queried via **AT+QIGETERROR**. Please note that **AT+QIGETERROR** just returns error code of the last TCP/IP AT command.

AT+QIGETERROR Query the Last Error Code

Test Command AT+QIGETERROR=?	Response OK
Execution Command AT+QIGETERROR	Response +QIGETERROR: <err>,<errcode_description> OK

Maximum Response Time	300 ms
Characteristics	/

Parameter

<err>	Integer type. Error code of operation. Please refer to Chapter 4 .
<errcode_description>	A string parameter indicates the details of error information. Please refer to Chapter 4 for details.

2.3. Description of URCs

The URC of TCP/IP AT commands will be reported to the host in the format of beginning with **+QIURC:**. It contains the reports about incoming data, connection closed and incoming connection and so on. Actually, there is **<CR><LF>** both before and after the URC, but **<CR><LF>** is not presented intentionally.

2.3.1. +QIURC: "closed" Indicate Connection Closed

When TCP socket service is closed by a remote peer or due to network error, the URC will be outputted, and the status of socket service will be "Closing" (**<socket_state>=4**). **AT+QICLOSE=<connectID>** can be used to change **<socket_state>** to "Initial".

+QIURC: "closed" Indicate Connection Closed

+QIURC: "closed",<connectID>	Socket service connection is closed.
------------------------------	--------------------------------------

Parameter

<connectID>	Integer type. Socket service index. Range: 0–11.
-------------	--

2.3.2. +QIURC: "recv" Indicate Incoming Data

In buffer access mode or direct push mode, after receiving data, the module will report URC to the host.

In buffer access mode, after receiving data, the module will report URC **+QIURC: "recv",<connectID>** to notify the host. Then the host can retrieve data via **AT+QIRD**. Please note that if the buffer is not empty, and the module receives data again, it will not report a new URC until all the received data has been retrieved via **AT+QIRD** from buffer.

In direct push mode, the received data will be outputted to COM port directly.

+QIURC: "recv" Indicate Incoming Data

+QIURC: "recv",<connectID>	Indicate incoming data in buffer access mode. The host can retrieve data via AT+QIRD .
+QIURC: "recv",<connectID>,<currentrecvlength><CR><LF><data>	Indicate incoming data in direct push mode when the <service_type> is "TCP", "UDP" or "TCP INCOMING".
+QIURC: "recv",<connectID>,<currentrecvlength>,<remoteIP>,<remote_port><CR><LF><data>	Indicate incoming data in direct push mode when <service_type> is "UDP SERVICE".

Parameter

<connectID>	Integer type. Socket service index. Range: 0–11.
<currentrecvlength>	Integer type. Length of actually received data.
<remoteIP>	String type. Remote IP address.
<remote_port>	Integer type. Remote port number.
<data>	String type. Received data.

2.3.3. +QIURC: "incoming full" Indicate Incoming Connection Full

If an incoming connection reaches the limit, or no socket system resources can be allocated, then the module will report the URC **+QIURC: "incoming full"** for the new incoming connection request.

+QIURC: "incoming full" Indicate Incoming Connection Full

+QIURC: "incoming full"	The URC indicates the incoming connection is full.
-------------------------	--

2.3.4. +QIURC: "incoming" Indicate Incoming Connection

If <service_type> is "TCP LISTENER", when a remote client connects to this server, the host will automatically assign a free <connectID> for the new connection, and the range of <connectID> is 0–11. In such a case, the module will report the URC **+QIURC: "incoming"**. The <service_type> of the new connection will be "TCP INCOMING", and the <access_mode> will be buffer access mode.

+QIURC: "incoming" Indicate Incoming Connection

+QIURC: "incoming",<connectID>,<serverID>,<remoteIP>,<remote_port>	When the new incoming connection is accepted by <serverID>, the allocated <connectID>, <remoteIP> and <remote_port> will be informed by this URC.
--	---

Parameter

<connectID>	Integer type. Assign this socket service for the incoming connection, which is automatically specified by the module. Range: 0–11.
<serverID>	Integer type. The incoming <connectID> accepted by the server whose <service_type> is "TCP LISTENER" and listening socket ID is <serverID> .
<remoteIP>	String type. Remote IP address of the incoming <connectID> .
<remote_port>	Integer type. Remote port number of the incoming <connectID> .

2.3.5. +QIURC: "pdpdeact" Indicate PDP Deactivation

PDP context may be deactivated by the network. The module will report this URC to the host about PDP deactivation. In such a case, the host must execute **AT+QIDEACT** to deactivate the context and reset all connections.

+QIURC: "pdpdeact" Indicate PDP Deactivation

+QIURC: "pdpdeact",<contextID>	<contextID> context is deactivated.
---	--

Parameter

<contextID>	Integer type. Context ID. Range: 1–16.
--------------------------	--

3 Examples

3.1. Configure and Activate a Context

3.1.1. Configure a Context

```
AT+QICSGP=1,1,"UNINET","", "",1 //Configure context 1. APN is "UNINET" for China Unicom.  
OK
```

3.1.2. Activate a Context

```
AT+QIACT=1 //Activate context 1. Depending on the network, the  
OK //Activated the context successfully.  
AT+QIACT? //Query the context state, protocol type and IP address of  
+QIACT: 1,1,1,"10.7.157.1" context 1.  
OK
```

3.1.3. Deactivate a Context

```
AT+QIDEACT=1 //Deactivate context 1.  
OK //Deactivated the context successfully. Depending on the  
network, the maximum response time is 40 s.
```

3.2. TCP Client Works in Buffer Access Mode under IPv4

3.2.1. Set up a TCP Client Connection and Enter Buffer Access Mode

```
AT+QIOPEN=1,0,"TCP","220.180.239.212",8009,0,0 //Context is 1 and <connectID> is 0. Before  
using AT+QIOPEN, the host should activate  
the context with AT+QIACT first.
```

OK

+QIOPEN: 0,0

//TCP client is connected successfully. It is suggested to wait for 150 seconds for the URC **+QIOPEN: <connectID>,<err>**. If the URC cannot be received in 150 seconds, the host can use **AT+QICLOSE** to close the socket.

AT+QISTATE=1,0

//Query the connection status of socket service 1.

+QISTATE: 0,"TCP","220.180.239.201",8705,65514,2,1,0,0,"usbmodem"

OK

3.2.2. Send Data in Buffer Access Mode

AT+QISEND=0

//Send variable-length data.

>test1<ctrl+Z>

SEND OK

//**SEND OK** does not mean the data has been sent to the server successfully. The host can query whether the data has reached the server via **AT+QISEND=0,0**.

AT+QISEND=0,4

//Send fixed-length data and the data length is 4 bytes.

>test

SEND OK

AT+QISEND=0,0

//Query the length of sent data, acknowledged data and unacknowledged data.

+QISEND: 9,9,0

OK

AT+QISENDEX=0,"3132333435"

//Send Hex string data.

SEND OK

AT+QISEND=0,0

//Query the length of the sent data, acknowledged data and unacknowledged data..

+QISEND: 14,14,0

OK

3.2.3. Retrieve Data from Remote Server in Buffer Access Mode

+QIURC: "recv",0

//<connectID> 0 has received data.

AT+QIRD=0,1500

//Retrieve the data. The maximum length of data to be retrieved is 1500 bytes.

```
+QIRD: 5 //The length of actually retrieved data is 5 bytes.
test1

OK
AT+QIRD=0,1500
+QIRD: 0 //No data in buffer.

OK
AT+QIRD=0,0 //Query the total length of received data, including read and unread data.
+QIRD: 5,5,0

OK
```

3.2.4. Close a Connection

```
AT+QICLOSE=0 //Close a connection whose <connectID> is 0. Depending on the
network, the maximum response time is 10 seconds.

OK
```

3.3. TCP Client Works in Buffer Access Mode under IPv6

3.3.1. Set up a TCP Client Connection and Enter Buffer Access Mode

```
AT+QIOPEN=1,0,"TCP","2001:468:2100:1:0:0:0:0",5001,0,0 //Context is 1 and <connectID> is 0.
Before using AT+QIOPEN, the host should
activate the context with AT+QIACT first.

OK

+QIOPEN: 0,0 //TCP client is connected successfully. It is
suggested to wait for 150 seconds for the
URC +QIOPEN: <connectID>,<err>. If the
URC cannot be received in 150 seconds,
the host can use AT+QICLOSE to close the
socket.

AT+QISTATE=1,0 //Query the connection status of socket
service 1.

+QISTATE: 0,"TCP","2001:468:2100:1:0:0:0:0",5001,11921,2,1,0,0,"usbmodem"

OK
```

3.3.2. Send Data in Buffer Access Mode

```

AT+QISEND=0 //Send variable-length data.
>test1<ctrl+Z>
SEND OK //SEND OK does not mean the data has been sent to the server
        //successfully. The host can query whether the data has reached the
        //server via AT+QISEND=0,0.

AT+QISEND=0,4 //Send fixed-length data and the data length is 4 bytes.
>test
SEND OK

AT+QISEND=0,0 //Query the length of sent data, acknowledged data and
                //unacknowledged data.

+QISEND: 9,9,0

OK

AT+QISENDEX=0,"3132333435" //Send Hex string data.
SEND OK

AT+QISEND=0,0 //Query the length of sent data, acknowledged data and
                //unacknowledged data.

+QISEND: 14,14,0

OK

```

3.3.3. Receive Data from Remote Server in Buffer Access Mode

```

+QIURC: "recv",0 //<connectID> 0 has received data.
AT+QIRD=0,1500 //Retrieve the data. The maximum length of data to be retrieved is 1500
                //bytes.

+QIRD: 5 //The length of actually retrieved data is 5 bytes.
test1

OK

AT+QIRD=0,1500
+QIRD: 0 //No data in buffer.

OK

AT+QIRD=0,0 //Query the total length of received data, including read and unread data.
+QIRD: 5,5,0

OK

```


3.3.4. Close a Connection

AT+QICLOSE=0

//Close a connection whose **<connectID>** is 0. Depending on the network, the maximum response time is 10 seconds.

OK

3.4. TCP Client Works in Transparent Access Mode

3.4.1. Set up a TCP Client Connection and Enter Transparent Access Mode

AT+QIOPEN=1,0,"TCP","220.180.239.212",8009,0,2

//Context is 1 and **<connectID>** is 0. Before using **AT+QIOPEN**, the host should activate the context with **AT+QIACT** first.

CONNECT

//TCP client connected successfully. It is suggested to wait for 150 seconds for the URC **CONNECT**. If the URC cannot be received in 150 seconds, the host could use **AT+QICLOSE** to close the socket.

3.4.2. Send Data in Transparent Access Mode

All data got from COM port will be sent to internet directly.

3.4.3. Receive Data from Remote Server in Transparent Access Mode

Test 1

//All data received from internet will be outputted via COM port directly.

3.4.4. Close a TCP Client

AT+QICLOSE=0

//After using **+++** to exit from the transparent access mode, the host could use **AT+QICLOSE** to close the TCP link. Depending on the network, the maximum response time is 10 seconds.

OK

3.5. TCP Client Works in Direct Push Mode

3.5.1. Set up a TCP Client Connection and Enter Direct Push Mode

```
AT+QIOPEN=1,0,"TCP","220.180.239.212",8009,0,1 //Context is 1 and <connectID> is 0. Before
                                                    using AT+QIOPEN, the host should activate the
                                                    context via AT+QIACT first.

OK

+QIOPEN: 0,0                                     //TCP client connected successfully. It is
                                                    suggested to wait for 150 seconds for the URC
                                                    +QIOPEN: <connectID>,<err>. If the URC
                                                    cannot be received in 150 seconds, the host
                                                    could use AT+QICLOSE to close the socket.

AT+QISTATE=1,0                                   //Query the connection state of socket service 0.
+QISTATE: 0,"TCP","220.180.239.201",8705,65344,2,1,0,1,"usbmodem"

OK
```

3.5.2. Send Data in Direct Push Mode

```
AT+QISEND=0                                     //Send variable-length data..
>test1<ctrl+Z>
SEND OK                                         //SEND OK does not mean the data has been sent to the
                                                    server successfully. The host can query whether the
                                                    data has reached the server via AT+QISEND=0,0.

AT+QISEND=0,5                                   //Send fixed-length data and the data length is 5 bytes.
>test2
SEND OK

AT+QISEND=0,0                                   //Query the length of sent data, acknowledged data and
                                                    unacknowledged data.
+QISEND: 10,10,0                               //A total of 10 bytes data has been sent, and all the
                                                    10-byte data has been acknowledged.

OK
```

3.5.3. Retrieve Data from Remote Server in Direct Push Mode

```
+QIURC: "recv",0,4                             //Retrieve data from remote server.
test
```

3.5.4. Close a TCP Client

AT+QICLOSE=0

//Close the connection whose **<connectID>** is 0.
Depending on the network, the maximum response time
is 10 seconds.

OK

3.6. TCP Server Works in Buffer Access Mode under IPv4

3.6.1. Start a TCP Server

AT+QIOPEN=1,1,"TCP LISTENER","127.0.0.1",0,2020,0 //Context is 1 and **<connectID>** is 1. Before
using **AT+QIOPEN**, the host should
activate the context with **AT+QIACT** first.

OK

+QIOPEN: 1,0

//TCP server is opened successfully.

AT+QISTATE=0,1

//Query whether the connection state of
<contextID> is 1.

+QISTATE: 1,"TCP LISTENER","10.7.157.1",0,2020,3,1,1,0,"usbmodem"

OK

3.6.2. Accept TCP Incoming Connection

+QIURC: "incoming",11,1,"172.31.242.222",54091 //Accept a TCP connection, **<service_type>** is
"TCP incoming", and **<connectID>** is 11.

3.6.3. Receive Data from Incoming Connection

+QIURC: "recv",11

//Receive data from remote incoming connection.

AT+QIRD=11,1500

//Retrieve the data received from incoming
connection.

+QIRD: 4

//Actually retrieved data length is 4 bytes.

test

OK

AT+QIRD=11,1500

+QIRD: 0

//No data in buffer.

```
OK
AT+QIRD=11,0 //Query the total length of retrieved data, including
               read and unread data.
+QIRD: 4,4,0
OK
```

3.6.4. Close a TCP Server

```
AT+QICLOSE=11 //Close the incoming connection. Depending on the
               network, the maximum response time is 10 seconds.
OK
AT+QICLOSE=1 //Close TCP server listening.
OK
```

3.7. TCP Server Works in Buffer Access Mode under IPv6

3.7.1. Start a TCP Server

```
AT+QIOPEN=1,1,"TCP LISTENER","0:0:0:0:0:0:1",0,12345,0 //Context is 1 and <connectID> is 1.
                                                         Before using AT+QIOPEN, the host should
                                                         activate the context with AT+QIACT first.
OK
+QIOPEN: 1,0 //TCP server is opened successfully.
AT+QISTATE=0,1 //Query whether the connection state of
               <contextID> is 1.
+QISTATE: 1,"TCP LISTENER","2001:468:2100:1:2001:468:2100:1",0,12345,3,1,1,0,"usbmodem"
OK
```

3.7.2. Accept TCP Incoming Connection

```
+QIURC: "incoming",11,1,"2001:468:2100:1:0:0:0:0",54091 //Accept a TCP connection,
                                                         <service_type> is "TCP incoming", and
                                                         <connectID> is 11.
```

3.7.3. Receive Data from Incoming Connection

```
+QIURC: "recv",11 //Receive data from remote incoming connection.
AT+QIRD=11,1500 //Retrieve the data received from incoming
                  connection.
+QIRD: 4 //Actually retrieved data length is 4 bytes.
test
OK
AT+QIRD=11,1500
+QIRD: 0 //No data in buffer.
OK
AT+QIRD=11,0 //Query the total length of received data, including
              read and unread data.
+QIRD: 4,4,0
OK
```

3.7.4. Close a TCP Server

```
AT+QICLOSE=11 //Close the incoming connection. Depending on the
               network, the maximum response time is 10 seconds.
OK
AT+QICLOSE=1 //Close TCP server listening.
OK
```

3.8. UDP Service under IPv4

3.8.1. Start a UDP Service

```
AT+QIOPEN=1,2,"UDP SERVICE","127.0.0.1",0,3030,0 //Start a UDP service, <connectID> is 2 and
                                                    <context> is 1. Before using AT+QIOPEN,
                                                    the host should activate the context with
                                                    AT+QIACT first.
OK
+QIOPEN: 2,0 //UDP service is opened successfully.
AT+QISTATE=0,1 //Query if the connection status of
                <contextID> is 1.
```

```
+QISTATE: 2,"UDP SERVICE","10.7.157.1",0,3030,2,1,2,0,"usbmodem"
```

```
OK
```

3.8.2. Send UDP Data to Remote

```
AT+QISEND=2,10,"10.7.89.10",6969
```

//Send 10 bytes data to remote whose IP is 10.7.89.10 and remote port is 6969.

```
>1234567890
```

```
SEND OK
```

3.8.3. Receive Data from Remote

```
+QIURC: "recv",2
```

```
AT+QIRD=2
```

//Receive data from remote.

//Read UDP data. One whole UDP packet will be outputted. There is no need to specify the read length.

```
+QIRD: 4,"10.7.76.34",7687
```

//Data length is 4. The remote IP address is 10.7.76.34 and remote port is 7687.

```
AAAA
```

```
OK
```

```
AT+QIRD=2
```

//Read data.

```
+QIRD: 0
```

//No data in buffer.

```
OK
```

```
AT+QISEND=2,10,"10.7.76.34",7687
```

//Send data to remote whose IP is 10.7.76.34 and remote port is 7687.

```
>1234567890
```

```
SEND OK
```

3.8.4. Close a UDP Service

```
AT+QICLOSE=2
```

//Close the service.

```
OK
```

3.9. UDP Service under IPv6

3.9.1. Start a UDP Service

```
AT+QIOPEN=1,2,"UDP SERVICE","0:0:0:0:0:0:1",0,12345,0 //Start a UDP service, <connectID>
                                                         is 2 and <context> is 1. Before using
                                                         AT+QIOPEN, the host should activate the
                                                         context with AT+QIACT first.

OK

+QIOPEN: 2,0 //UDP service is opened successfully.
AT+QISTATE=0,1 //Query if the connection status of
                                                         <contextID> is 1.

+QISTATE: 2,"UDP SERVICE","2001:468:2100:1:2001:468:2100:1",0,12345,2,1,2,0,"usbmodem"

OK
```

3.9.2. Send UDP Data to Remote

```
AT+QISEND=2,10,"2001:468:2100:1:0:0:0:0",5001 //Send 10 bytes data to remote whose IP is
                                                         2001:468:2100:1:0:0:0:0 and remote port is
                                                         5001.

>1234567890
SEND OK
```

3.9.3. Receive Data from Remote

```
+QIURC: "recv",2 //Receive data from remote.
AT+QIRD=2 //Read UDP data. One whole UDP packet will
                                                         be outputted. There is no need to specify the
                                                         read length.

+QIRD: 4,"2001:468:2100:1:0:0:0:0",5001 //Data length is 4. The remote IP address is
                                                         2001:468:2100:1:0:0:0:0 and remote port is
                                                         5001.

AAAA

OK
AT+QIRD=2 //Read data.
+QIRD: 0 //No data in buffer.

OK
```

3.10. Ping a Remote Server

AT+QPING=1,"www.baidu.com"

//Ping www.baidu.com in context 1. Before pinging the destination IP address, the host should activate the context by **AT+QIACT** first.

OK

+QPING: 0,"61.135.169.125",32,192,255

+QPING: 0,"61.135.169.125",32,240,255

+QPING: 0,"61.135.169.125",32,241,255

+QPING: 0,"61.135.169.125",32,479,255

+QPING: 0,4,4,0,192,479,287

AT+QPING=1,"2001:468:2100:1:0:0:0:0"

//Ping 2001:468:2100:1:0:0:0:0 in context 1. Before pinging the destination IP address, the host should activate the context by **AT+QIACT** first.

OK

+QPING: 0,"2001:468:2100:1:0:0:0:0",32,58,255

+QPING: 0,"2001:468:2100:1:0:0:0:0",32,49,255

+QPING: 0,"2001:468:2100:1:0:0:0:0",32,39,255

+QPING: 0,"2001:468:2100:1:0:0:0:0",32,40,255

+QPING: 0,4,4,0,39,58,46

3.11. Synchronize Local Time

AT+QNTTP=1,"202.120.2.101",123

//Synchronize local time with NTP server "202.120.2.101:123". Before synchronizing the time, the host should activate the context with **AT+QIACT** first.

OK

+QNTTP: 0,"2013/08/19,11:10:10+32"

AT+CCLK?

+CCLK: "13/08/19,11:10:57+32"

OK

3.12. Get Last Error Code

AT+QIOPEN=1,"TCP","220.180.239.212",8009,0,1 //Start a socket service, but <connectID> is not specified.

ERROR

AT+QIGETERROR

+QIGETERROR: 552, invalid parameters

OK

4 Summary of Error Codes

If **<err>** is returned after executing TCP/IP AT commands, the details of error can be queried via **AT+QIGETERROR**. Please note that **AT+QIGETERROR** just returns error code of the last TCP/IP AT command.

Table 2: Summary of Error Codes

<err>	Description
0	Operation successful
550	Unknown error
551	Operation blocked
552	Invalid parameters
553	Memory not enough
554	Create socket failed
555	Operation not supported
556	Socket bind failed
557	Socket listen failed
558	Socket write failed
559	Socket read failed
560	Socket accept failed
561	Open PDP context failed
562	Close PDP context failed
563	Socket identity has been used
564	DNS busy

565	DNS parse failed
566	Socket connect failed
567	Socket has been closed
568	Operation busy
569	Operation timeout
570	PDP context broken down
571	Cancel send
572	Operation not allowed
573	APN not configured
574	Port busy

5 Appendix A Reference

Table 3: Terms and Abbreviations

Abbreviation	Description
3GPP	3rd Generation Partnership Project
ACK	Acknowledge
APN	Access Point Name
CHAP	Challenge Handshake Authentication Protocol
DNS	Domain Name System
FIN	Finish
ID	Identify
IP	Internet Protocol
IPv4	Internet Protocol Version 4
IPv6	Internet Protocol Version 6
NTP	Network Time Protocol
NVM	Non-Volatile Memory
PAP	Password Authentication Protocol I
PDP	Packet Data Protocol
PPP	Point-to-Point Protocol
QoS	Quality of Service
TCP	Transmission Control Protocol
UART	Universal Asynchronous Receiver & Transmitter
UDP	User Datagram Protocol

URC	Unsolicited Result Code
USB	Universal Serial Bus
UTC	Coordinated Universal Time
